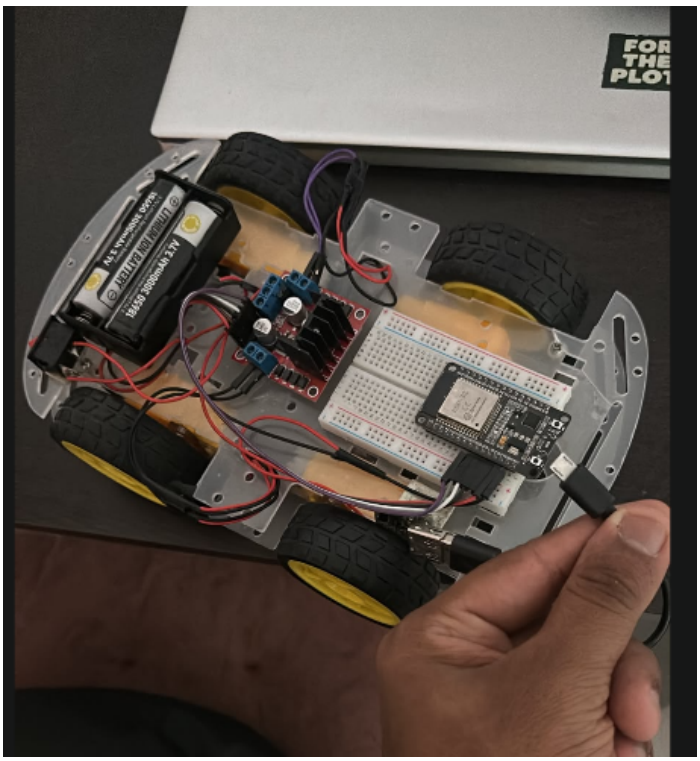
SERBOT | ESP32 Wi-Fi Robot Car

Working principle, wiring, operating instructions and troubleshooting



1. How it works

Phone / laptop → Wi-Fi hotspot created by ESP32 → **ESP32 GPIO commands** → **L298N motor driver** → DC motors and wheels.

The ESP32 creates its own wireless network, so an internet connection or Blynk is not required. The web dashboard sends movement commands to the ESP32, which changes the motor-driver input pins.

2. Start and operate the car

1. Upload the supplied SerBot program using Arduino IDE (board: ESP32 Dev Module; select the correct port).

2. Ensure ESP32 and L298N are correctly powered, with a common GND.

3. On your phone, join Wi-Fi **RoboCar**, password **robot123**. Stay connected even if the phone warns that there is no internet.

4. Open a browser and enter **http://192.168.4.1**.

5. Use the on-screen buttons. Test with the wheels raised off the surface first.

3. Controls

Dashboard button	What the car does
FWD (W)	Both motor channels drive forward
BACK (S)	Both motor channels drive backward

Dashboard button	What the car does
LEFT (A)	Left side stops; right side drives forward
RIGHT (D)	Right side stops; left side drives forward
STOP	All motors stop

Note: In the supplied code, LEFT and RIGHT only steer while the robot is already moving forward. Releasing a control sends STOP.

4. Wiring: L298N to ESP32

L298N pin	ESP32 pin	Purpose
IN1	GPIO 26	Left motor direction
IN2	GPIO 27	Left motor direction
IN3	GPIO 14	Right motor direction
IN4	GPIO 12	Right motor direction
ENA	GPIO 25	Left motor channel enable
ENB	GPIO 13	Right motor channel enable (your wiring)
GND	GND	Shared electrical ground

Required code edit: Change `#define ENB 15` to `#define ENB 13` and upload again. All other pin definitions above match the supplied program.

5. Power and safety

Supply the L298N motor-power input from a suitable battery pack for the motors. Power the ESP32 from USB or a correctly regulated supply. **Never feed raw motor-battery voltage into the ESP32 3.3V pin.** Connect grounds together. Turn power off before moving wires. Secure loose wiring before driving.

GPIO 12 is an ESP32 strapping pin: if the board fails to boot with the driver attached, move IN4 to GPIO 13 or GPIO 33 **only if that pin is free**, and update the pin definition. Because your ENB already uses GPIO 13, use GPIO 33 for IN4 if relocation is needed.

6. Troubleshooting

RoboCar Wi-Fi not visible: Check ESP32 power, upload success and serial output at 115200 baud.

Page at 192.168.4.1 does not load: Join RoboCar Wi-Fi; disable automatic switching to mobile data if necessary.

Dashboard works, wheels do not move: Check motor battery, common ground, ENA = GPIO 25, ENB = GPIO 13, and L298N outputs.

Forward moves backward: Motor leads / sides may be reversed; inspect wiring before changing code.

ESP32 resets or fails to start: Check supply regulation, motor noise and GPIO 12 boot-strapping issue.

7. Additional dashboard functions

The uploaded SerBot version logs command round-trip time (RTT), Wi-Fi signal strength (RSSI), packet loss, latency statistics, and speed/turn trial calculations. Logged data can be exported as CSV. These are measurements and

calculators; the car has no automatic wheel-speed sensor in the supplied setup.